

# Electron Orbits from the Signed Electrostatic Force: Covalent, Metallic, and Intermolecular Magnitudes

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## Abstract

On the equal-scale locus the electromagnetic force of double spacetime theory is the Coulomb force divided by the interference factor  $\gamma_s$ . The sign of  $\gamma_s$  is the repulsion. An unlike pair attracts where  $\gamma_s > 0$  and repels where  $\gamma_s < 0$ ; a like pair does the opposite. No centrifugal term is added to cancel an attraction. A single electron–proton pair then has no force-free radius: the radial force is nonzero wherever  $\gamma_s$  is finite. Two electrons on a diameter, and eight at the vertices of a cube, each have exactly one radial balance in the first repulsive shell; the diameter is an angular maximum, and the counts 2, 8, and 18 are two chiral seats on the modes through degree 2, not indices of the nodes. Those restoring radii lie where  $\gamma_s < 0$ . In the outer well, where the nucleus attracts, a cube of charge  $Z \geq 3$  has no radial root: every chord is longer than the radius, and the sum of the outward coefficients lies between 2 and 3. A regular dodecahedron has a chord shorter than the radius. For every nuclear charge  $Z \geq 8$ , the response of its twenty vertices equals  $Z$  before that chord meets the first node. Two protons and two electrons do, and in the first repulsive shell they do so at exactly one square. Protons sit on one diagonal and electrons on the other, and the electrostatic potential has a positive-definite Hessian there. Like charges attract along the diagonals and unlike charges repel along the sides. The diagonal is  $\sqrt{2}$  times the side, the diagonal force is  $4.3532 ke^2/\ell^2$ , and the stiffness of the diagonal is  $8.178 ke^2/\ell^3$ . A nearest-neighbor bond that carries only the fraction  $2/z$  of that pair has further points where every electrostatic component vanishes. For eight and for twelve neighbors those points are saddles, and the partial forces stand in the ratios 0.04323 and 0.1704 to the covalent diagonal. Two neutral pairs a few side lengths apart keep a force after the monopole cancels. Identifying the side with the Bohr radius makes the diagonal  $\sqrt{2}a_0$ , against the Born–Oppenheimer separation  $1.4011 a_0$  of  $H_2$ . The same layered sign, read in the gravitational channel, is the exclusion of a shared seat: two electron masses placed inside the first gravitational node meet a simple pole. The smallness of  $Gm_e^2$  thins that neighborhood and does not remove the pole. A restoring radius is a sign boundary of the outward force, and the electron oscillates across it. With the side identified as  $a_0$ , the charge-8 dodecahedron oscillates at 196.8 nm and the covalent diagonal, on the proton mass, at  $4.451 \mu\text{m}$ . Ordinary temperature meets the infrared of the second. The electron quanta lie in the ultraviolet and remain empty. On the square the ratio of those wavelengths is the square root of the ratio of  $K/m$ , fixed before the side is identified with  $a_0$ . A transition frequency is an eigenvalue difference and cannot scale with temperature. The heat-ray peak scales as  $k_B T$ , and the temperature selects the root. Wien’s displacement is not derived.

## 1 Introduction

The electron-shell section of double spacetime theory writes an electrostatic force whose denominator is the scalar interference factor  $\gamma_s$  [1]. At large separation the factor tends to one and the Coulomb law is recovered. Closer in,  $\gamma_s$  passes through zero on a discrete set of radii and is negative between those nodes. A node is a divergence of the force. The radial potential on the outer side of a node is unbounded below.

A purely attractive Coulomb force does not vanish at any finite radius. The classical circular orbit cancels it by a centrifugal term  $L^2/(mr^3)$ , the kinematics of a conserved angular momentum. That term is not added here. The outward contribution is already in the signed force. Where  $\gamma_s$  is negative an unlike pair repels, and where  $\gamma_s$  is positive a like pair repels. An electron orbit, in this note, is a locus on which these electrostatic contributions cancel. It is not a balance between Coulomb attraction and a centrifugal barrier.

The note computes that locus for a two-center covalent arrangement, for a nearest-neighbor bond that carries only  $2/z$  of an electron pair, and for the residual force between two neutral pairs. A further stability

belongs to the gravitational channel. The main text reads the Pauli principle as the first repulsive layer of that channel, generated when two fermions overlap, not as a separate statistical postulate. Applied to electrons, the same layer expels two masses from one orbital seat. The electron mass is small, and the force that expels them is a pole. The equal-scale identification  $\alpha = \beta = \ell/r$  is the reading already used for the nuclear and Coulomb layers. The length  $\ell$  is not derived. Bohr's  $n^2$  rule and the fine-structure constant are not derived. The same refusal covers the shell counts: 2, 8, and 18 are two chiral seats on the monopole, the dipole, and the quadrupole, and a one-center diameter or cube closes only its radius, and only in the reversed layer. The laboratory well, in which the nucleus attracts, closes a dodecahedron of twenty vertices once the nuclear charge is at least 8. The restoring radius is also a sign boundary. Section 11 reads the radial oscillation across it, and compares the electron frequency with the infrared of a warm body. The frequency is  $\omega^2 = Kke^2/(m\ell^3)$ : one signed balance, and a separate inertia for each coordinate that moves. A transition frequency is an eigenvalue difference and contains no temperature. The heat-ray peak scales as  $k_B T$ , which contradicts that fixing, and the temperature in the signed force selects the root along the order of  $K/m$ . Where a laboratory number enters, it is marked as an input.

## 2 The signed force

On one axis the interference factor of the dual rotor is

$$\gamma_s = \cosh \frac{\alpha}{2} \cos \frac{\beta}{2} - \sinh \frac{\alpha}{2} \sin \frac{\beta}{2}.$$

The equal-scale locus sets  $\alpha = \beta = \ell/r$ . In the dimensionless variable  $x = \ell/(2r)$ ,

$$\gamma_s(x) = \cosh x \cos x - \sinh x \sin x, \quad \gamma'_s(x) = -2 \cosh x \sin x. \quad (1)$$

Thus  $\gamma_s(0) = 1$ , and on each interval  $[n\pi, (n+1)\pi]$  there is one simple zero. The first zero  $x_1$  is the unique root of

$$\tan x = \coth x \quad (2)$$

in  $(\pi/4, \pi/2)$ . Numerically  $x_1 = 0.93755$ , so the outermost node lies at

$$\frac{r_1}{\ell} = \frac{1}{2x_1} = 0.53330.$$

The second zero lies at  $x_2 = 3.9274$ , and the inner edge of the first repulsive shell is  $r_2/\ell = 0.12731$ . Ordered by decreasing radius, the outer well  $r > r_1$  has  $\gamma_s > 0$ , the shell  $r_2 < r < r_1$  has  $\gamma_s < 0$ , and the sign continues to alternate. The extremum inside the first shell is  $\gamma_s(\pi) = -\cosh \pi$ , at  $r = \ell/(2\pi)$ .

The force that increases the separation of two charges is

$$F_r = \frac{kq_1q_2}{\gamma_s(r)r^2}, \quad k = \frac{1}{4\pi\epsilon_0}. \quad (3)$$

Like charges repel where  $\gamma_s > 0$  and attract where  $\gamma_s < 0$ . Unlike charges do the opposite. Both directions are the same formula. Nothing further is added to produce a repulsion.

The Taylor expansion at the origin follows from  $\gamma''_s(0) = -2$  and  $\gamma''''_s(0) = -4$ ,

$$\gamma_s(x) = 1 - x^2 - \frac{1}{6}x^4 + O(x^6). \quad (4)$$

Hence

$$\frac{1}{\gamma_s(r)r^2} = r^{-2} + \frac{\ell^2}{4}r^{-4} + \frac{7\ell^4}{96}r^{-6} + O(r^{-8}).$$

The correction to the Coulomb force begins at order  $(\ell/r)^2$ .

For  $r > r_1$  the pair potential that vanishes at infinity is the integral of (3). The substitution  $s = \ell/(2x)$  gives  $ds/s^2 = -(2/\ell)dx$ , and therefore

$$\int_r^\infty \frac{ds}{\gamma_s(s)s^2} = \frac{2}{\ell} \int_0^{\ell/(2r)} \frac{dx}{\gamma_s(x)}, \quad (5)$$

provided the path stays in the outer well. The integral diverges as the upper limit approaches  $x_1$ . Inside a shell the same formula does not continue: the path from infinity would cross a pole. A local force balance inside the shell is still well defined, because (3) requires only  $\gamma_s \neq 0$ .

### 3 No centrifugal term

Consider one proton and one electron. The radial force is

$$F_r = -\frac{ke^2}{\gamma_s(r)r^2}.$$

Wherever  $\gamma_s$  is finite and nonzero,  $F_r$  is nonzero. In the outer well it is attractive at every finite radius. In the first repulsive shell it is repulsive at every finite radius. There is no radius at which the electrostatic force vanishes, and none is introduced by hand.

The centripetal rearrangement  $v^2 = ke^2/(m\gamma_s r)$  of the main paper records what a conserved angular momentum would contribute in the outer well. It is a kinematic identity for a circular Newtonian orbit. It is not an extra force in (3), and it is not used below. The repulsive shell already reverses the electron–proton force. At its quietest point,  $x = \pi$ ,

$$|F| = \frac{4\pi^2}{\cosh \pi} \frac{ke^2}{\ell^2} = 3.4057 \frac{ke^2}{\ell^2}, \quad (6)$$

directed outward. That is the built-in repulsion, evaluated where  $|\gamma_s|$  in the first shell is largest and the force is therefore smallest. A one-center electron is not parked on a force-free circle. A force-free locus appears only when several charges are present and their signed contributions cancel.

### 4 Shell counts

The integers 2, 8, and 18 are not labels of the equal-scale nodes. Those nodes still march inward as  $\Theta(1/n)$ . Bohr’s ladder of radii  $n^2$  is not read off from them.

Two electrons at the ends of a diameter, about a nucleus of charge  $Z \geq 1$ , have no radial balance in the outer well. In the first repulsive shell, where both the radius and the diameter lie between the first node and the second, there is exactly one, whichever  $Z$  is. There the nucleus pushes outward and the two electrons attract. On the side of larger radius the outward force points inward, and on the side of smaller radius it points outward, so the radius is restored. The angle is not. At a fixed radius the separation of the pair is greatest when the electrons are antipodal, and wherever the interference factor of that separation is negative an energy that increases with separation has a maximum there. A displacement that draws the two electrons toward one common seat is not restored by the electrostatic force.

Eight electrons at the vertices of a cube about the same nucleus have, again for every  $Z \geq 1$ , exactly one radial balance in the window where the radius and the three separations—the edge, the face diagonal, and the body diagonal—all lie in the first repulsive shell. The outward force restores that radius.

Eighteen is not the number of vertices of one cubic orbit. Those orbits have six, eight, or twelve. Eighteen counts seats. A monopole is one mode, a dipole three, and a traceless quadrupole five. Two complementary chiralities put one seat on each mode. Through degree zero the count is 2, through degree one it is 8, and through degree two it is 18. The same arithmetic is  $2n^2$  for  $n = 1, 2, 3$ . It does not locate a node, and no eighteen-point figure is claimed here to be a minimum of the force.

The diameter’s slide toward a single seat is the overlap of Section 6. The gravitational pole expels it. The locus on which the signed forces cancel, and the Hessian is positive, is the two-center square.

### 5 The laboratory well

The sign that scattering and spectroscopy measure at atomic distances is the sign of the outer well. There  $\gamma_s > 0$ : a nucleus attracts an electron, electrons repel one another, and a path from any finite radius out to infinity does not cross a node. The integral (5) therefore assigns a finite removal energy. The reversed layer does the opposite. Like charges attract, unlike charges repel, and the path to infinity is cut by the pole.

The restoring radii of Section 4 lie in that reversed layer. They are not ionization equilibria. For the cube the obstruction is geometric and holds for every nuclear charge  $Z \geq 3$ . Each of the three chords—the edge, the face diagonal, and the body diagonal—is longer than the radius, so each electron–electron phase is smaller than the nuclear phase. On  $(0, x_1)$  the interference factor is strictly decreasing, and the response built from those three chords therefore lies strictly below its far-field value. That value is the sum of the outward coefficients,

$$\frac{3\sqrt{3}}{4} + \frac{3\sqrt{6}}{8} + \frac{1}{4},$$

which lies strictly between 2 and 3. For  $Z \geq 3$  the response never meets the charge, and the outward force remains inward throughout the outer well. A neutral cube, and every heavier one, does not close where the laboratory signs hold. The same is already true of the diameter for every  $Z \geq 1$ : the single coefficient is  $1/4$ .

A restoring radius appears in the outer well when some like-charge chord is shorter than the radius. Shrinking the figure then drives that chord into the first node while the radius is still outside it, and the pair repulsion diverges outward. At large separation the same figure, if its geometric coefficient lies below the nuclear charge, falls inward. Between those two regimes the response crosses the charge.

The regular dodecahedron has such a chord. From each vertex, three neighbors lie at squared distance  $2(3 - \sqrt{5})/3$  times the squared radius, so the scale  $R/d$  of that chord is greater than one. The far-field coefficient of the twenty vertices is strictly less than 8. For every nuclear charge  $Z \geq 8$ , the response—the sum of the chord coefficients times  $\gamma_s$  at the radius divided by  $\gamma_s$  at the chord—therefore equals  $Z$  at a phase  $x$  strictly before the chord meets the first node. At that phase every pair is still in the outer well, the radial force vanishes, and the removal to infinity is the integral (5). As the separation tends to infinity the response tends to the same coefficient, which is less than  $Z$ , so the far field is inward. Beside the chord node the response exceeds every bound, so the force is outward.

Twenty is the number of vertices, not a principal shell. The counts 2, 8, and 18 remain two chiral seats on the modes. A shell of eight, packed as a cube, still closes only in the reversed layer. The laboratory closure is the one whose shortest like-chord meets the node first.

## 6 Exclusion as gravitational repulsion

The gravitational channel uses the same interference factor, sourced by the kinematic pair tied to inertial mass rather than by the pair tied to charge [1]. For two electron masses the force that increases their separation is

$$F_G = -\frac{Gm_e^2}{\gamma_{s,G}(r)r^2}, \quad (7)$$

with  $\gamma_{s,G}$  the equal-scale factor of that channel and with its own length  $\ell_G$ . Where  $\gamma_{s,G} > 0$  the masses attract. Where  $\gamma_{s,G} < 0$  they repel. The first zero of  $\gamma_{s,G}$  is simple, by the same derivative (1). Approached from inside,  $F_G$  therefore has a simple pole and points outward.

The prefactor is small. The CODATA values give

$$\frac{Gm_e^2}{ke^2} = 2.401 \times 10^{-43}. \quad (8)$$

A simple pole is not removed by a small prefactor. Write the inward fractional displacement from the first gravitational node as  $r = r_1(1 - \varepsilon)$  with  $\varepsilon \downarrow 0$ . Then  $x - x_1 = x_1\varepsilon + O(\varepsilon^2)$  and

$$|\gamma_{s,G}| = 2x_1 \cosh x_1 \sin x_1 \varepsilon + O(\varepsilon^2) = 2.2260 \varepsilon + O(\varepsilon^2).$$

The outward force exceeds the scale  $ke^2/r^2$  of a nonsingular electromagnetic force at the same radius when  $\varepsilon < 1.078 \times 10^{-43}$ . The electron mass fixes the width of that neighborhood. It does not decide whether the neighborhood exists.

This comparison assumes that the electromagnetic factor is not itself zero at the same radius. The two channels are distinct phase pairs, so a gravitational node is not required to be an electromagnetic node, and  $\ell_G$  is not fixed by the square of the next section. If the two factors vanished together, their ratio would remain (8) and the electromagnetic piece would dominate. The exclusion read here is the gravitational pole at a node the charge channel does not share.

Placing both electrons on one orbital seat drives their separation through that node. The force (7) is then outward and unbounded, so the shared seat is left at once. The main text proposes that this repulsive layer is the Pauli principle: fermions entering it cannot occupy one seat, without a prior appeal to quantum statistics. That proposal is adopted here for electrons. It is not a derivation of the spin-statistics theorem, and it does not fix  $\ell_G$ .

## 7 The square and the covalent force

Place protons of charge  $+e$  at  $z = \pm R/2$  and electrons of charge  $-e$  at cylindrical radius  $\rho$ , opposite each other in the midplane. The three separations are the proton–proton distance  $R$ , the electron–electron

distance  $r_{ee} = 2\rho$ , and the four equal electron–proton distances

$$r_{ep} = \sqrt{(R/2)^2 + \rho^2}.$$

No angular momentum is assigned. Equilibrium means that the axial force on each proton and the cylindrical force on each electron both vanish.

The axial force on the proton at  $+R/2$ , counted positive when it increases  $R$ , and the cylindrical force on one electron, counted positive when it increases  $\rho$ , are

$$F_z = ke^2 \left( \frac{1}{\gamma_s(R) R^2} - \frac{R}{\gamma_s(r_{ep}) r_{ep}^3} \right), \quad (9)$$

$$F_\rho = ke^2 \left( \frac{1}{\gamma_s(r_{ee}) r_{ee}^2} - \frac{2\rho}{\gamma_s(r_{ep}) r_{ep}^3} \right). \quad (10)$$

The first term of  $F_\rho$  is the electron–electron force. In the outer well it is a repulsion and points outward: that outward piece is electrostatic, and it occupies the place a centrifugal term would have occupied. In the repulsive shell  $\gamma_s(r_{ee}) < 0$ , so the same term points inward, because like charges attract there.

The configuration in which the four charges are the vertices of a square, with the protons on one diagonal and the electrons on the other, is

$$\rho = \frac{R}{2}, \quad r_{ee} = R, \quad r_{ep} = \frac{R}{\sqrt{2}}. \quad (11)$$

On this locus  $F_z$  and  $F_\rho$  reduce to one expression. With  $x = \ell/(2R)$ ,

$$F_z = F_\rho = \frac{ke^2}{R^2} \left( \frac{1}{\gamma_s(x)} - \frac{2\sqrt{2}}{\gamma_s(\sqrt{2}x)} \right). \quad (12)$$

Both vanish when

$$\gamma_s(\sqrt{2}x) = 2\sqrt{2}\gamma_s(x). \quad (13)$$

The coefficient  $2\sqrt{2}$  is greater than one, and on  $[0, \pi]$  the factor  $\gamma_s$  decreases. While the diagonal phase  $x$  is still less than the first node,  $\gamma_s(x)$  is positive and  $\sqrt{2}x$  has not yet reached  $\pi$ . The value at the larger phase is then smaller than the value at  $x$ , and that value is smaller than  $2\sqrt{2}$  times itself. The two sides of the balance do not meet. They still do not meet if only the side has crossed the node, for then one factor is negative and the other is positive. A square whose diagonal lies in the outer well is not an equilibrium.

The balance can hold only after the diagonal has entered the first repulsive shell, so that both the diagonal and the side lie between the two nodes. There both factors are negative: like charges attract along the diagonals, and unlike charges repel along the sides. On that range

$$\gamma_s(\sqrt{2}x) - 2\sqrt{2}\gamma_s(x)$$

increases with  $x$ . It is negative at  $3/2$  and positive at  $8/5$ , so it vanishes once, and only once, in the shell. The crossing lies between those two phases. At  $8/5$  the side phase is still less than  $\pi$ , and the second node lies beyond  $\pi$ , so the crossing is inside the shell. Its phase is

$$x = 1.55463, \quad \frac{R}{\ell} = 0.32162, \quad \frac{r_{ep}}{\ell} = 0.22742.$$

The two interference factors are  $\gamma_s(x) = -2.2208$  on the diagonals and  $\gamma_s(\sqrt{2}x) = -6.2812$  on the sides. The repulsion that holds the electrons off the protons is the side force. It is not a centrifugal force.

The partial forces at this root are

$$|F_{\text{diag}}| = \frac{ke^2}{|\gamma_s(x)| R^2} = 4.3532 \frac{ke^2}{\ell^2}, \quad (14)$$

$$|F_{\text{side}}| = \frac{ke^2}{|\gamma_s(\sqrt{2}x)| r_{ep}^2} = 3.0782 \frac{ke^2}{\ell^2}. \quad (15)$$

The diagonal force is the covalent force scale: it is the attractive force along the proton–proton separation, cancelled by the axial part of the side repulsions.

The electrostatic potential, regarded as a function of  $R$  and  $\rho$  near this root, has Hessian eigenvalues  $37.33 ke^2/\ell^3$  and  $7.933 ke^2/\ell^3$ , both positive. The root is a local minimum. Allowing  $\rho$  to follow  $R$ , the stiffness of the diagonal is

$$\kappa = 8.178 \frac{ke^2}{\ell^3}. \quad (16)$$

Two further roots of  $F_z = F_\rho = 0$  lie in the same shell. Each has Hessian eigenvalues of opposite sign, so each is a saddle. The square is the stable covalent orbit. Its two electrons occupy opposite vertices, a separation  $R$ , not one seat. A displacement that puts both on one vertex is the overlap of Section 6: once that separation reaches the first gravitational node,  $F_G$  expels them. The electromagnetic Hessian stabilizes the square against small displacements. The gravitational pole forbids the shared seat even though  $m_e$  is small. Because the root lies inside the shell, the outer-well integral (5) does not assign it a dissociation energy relative to separated atoms at infinity.

## 8 Coordination and the metallic force

A nearest-neighbor bond in a monovalent coordination shell of  $z$  neighbors is given the electron content  $\nu = 2/z$ . The dimer is  $z = 1$ ,  $\nu = 2$ , the square of the previous section. The charge  $-\nu e$  is split into two lumps of charge  $-(\nu/2)e$  at opposite points of the midplane. The forces (9) and (10) become

$$F_z = ke^2 \left( \frac{1}{\gamma_s(R) R^2} - \frac{\nu R}{2 \gamma_s(r_{ep}) r_{ep}^3} \right), \quad (17)$$

$$F_\rho = ke^2 \left( \frac{(\nu/2)^2}{\gamma_s(r_{ee}) r_{ee}^2} - \frac{\nu \rho}{\gamma_s(r_{ep}) r_{ep}^3} \right). \quad (18)$$

Equilibrium is still  $F_z = F_\rho = 0$ , with no centrifugal term. The outward piece in the outer well is the first term of  $F_\rho$ : the lumps repel where  $\gamma_s(r_{ee}) > 0$ .

For  $z = 8$  and  $z = 12$  that outer-well system has one root each. All three separations lie outside  $r_1$ , so the signs are the ordinary ones: like charges repel, unlike charges attract. The partial proton–proton force and its ratio to the covalent diagonal (14) are

$$z = 8 : \quad \frac{R}{\ell} = 2.3591, \quad F_{\parallel} = 0.18820 \frac{ke^2}{\ell^2}, \quad \frac{F_{\parallel}}{|F_{\text{diag}}|} = 0.04323, \quad (19)$$

$$z = 12 : \quad \frac{R}{\ell} = 1.2666, \quad F_{\parallel} = 0.74188 \frac{ke^2}{\ell^2}, \quad \frac{F_{\parallel}}{|F_{\text{diag}}|} = 0.1704. \quad (20)$$

At both roots the Hessian of the potential has eigenvalues of opposite sign. Each is a saddle, not a local minimum. The same is true of the further roots that sit inside the repulsive shell. Reducing the electron content from a full pair to  $2/z$  leaves a point where the signed forces cancel, and it does not leave a stable orbit. A crystal in which each ion is surrounded symmetrically can still have a vanishing net force by symmetry. That lattice sum is not the calculation above. An empirical nearest-neighbor distance would fix  $F_{\parallel}$  by the proton–proton term alone; it is an input, and it is not required for the ratios.

## 9 Neutral pairs and the intermolecular force

The stable covalent orbit fixes an electron–proton distance  $a = R/\sqrt{2} = 0.22742 \ell$ . A neutral pair is a proton with an electron displaced by that distance. It is not a force-free one-center orbit: Section 3 has none. It is the side of the square, used as the size of a neutral pair. Two such pairs whose centers are a distance  $D$  apart have four intermolecular forces, each of the form (3). If each electron were coincident with its proton, every pair would share one distance and the sum would vanish with the net charge. The monopole cancels. What remains is the displacement  $a$ , read with the signed factor  $\gamma_s$ .

Two collinear placements are compared. In the parallel placement both electrons are displaced by  $+a$  along the line of centers. In the inward placement the electrons lie between the protons. Attraction is counted positive.  $F$  is the net force on the left-hand pair and  $F_C$  is the same sum at  $\gamma_s = 1$ , so that the difference is the torsional excess. In units of the covalent diagonal,

| $D/a$ | $F/ F_{\text{diag}} $ | $F_C/ F_{\text{diag}} $ | $F/ F_{\text{diag}} $ | $F_C/ F_{\text{diag}} $ |
|-------|-----------------------|-------------------------|-----------------------|-------------------------|
|       | parallel              |                         | inward                |                         |
| 4     | +0.5937               | +0.1160                 | +4.537                | −0.4010                 |
| 6     | +0.03668              | +0.02155                | −0.1056               | −0.04565                |
| 8     | +0.008803             | +0.006679               | −0.01666              | −0.01149                |

At four side lengths the inward electron–electron separation is  $2a = 0.4548\ell$ , which lies inside  $r_1$ . That pair has  $\gamma_s < 0$ , so the two electrons attract, and the net inward force is several times the covalent diagonal while the pure Coulomb sum is repulsive. The amplification is the sign change in (3), not an extra interaction. At six and eight side lengths every intermolecular separation is outside the node. Both columns fall. At eight side lengths the parallel residue is 0.88% of the covalent diagonal, and about a quarter of that residue is the torsional excess over  $F_C$ .

## 10 A numerical illustration

The ratio of diagonal to side is  $\sqrt{2}$  by (11), with no choice of  $\ell$ . One identification turns the side into the Bohr radius,

$$a = r_{ep} = a_0.$$

The diagonal is then

$$R = \sqrt{2} a_0 = 1.4142 a_0,$$

against the Born–Oppenheimer equilibrium separation  $1.4011 a_0$  of  $\text{H}_2$  [3]. The predicted separation is long by a factor 1.009. The identification  $a = a_0$  is an input; the factor  $\sqrt{2}$  is the square. With the CODATA values of  $a_0$  and of the Hartree energy  $E_h = ke^2/a_0$  [2], the same identification gives  $\ell = 4.3971 a_0$  and

$$|F_{\text{diag}}| = 18.55 \text{ nN}, \quad |F_{\text{side}}| = 13.12 \text{ nN}, \quad \kappa = 150 \text{ N m}^{-1}.$$

The diagonal force is the attractive proton–proton force at the equilibrium. The side force is the repulsive electron–proton force. The stiffness is (16). No dissociation energy is quoted: the equilibrium lies in the repulsive shell, and (5) does not cross the node. The measured dissociation energy of ortho- $\text{H}_2$  from  $N = 1$  is 4.463 eV [4]; it is an external number, and it is not a value of this potential.

## 11 Vibration across the sign boundary

A restoring radius is the boundary between an inward force and an outward force. On the side of larger radius the outward force points inward, and on the side of smaller radius it points outward. A displacement off the root oscillates across that change of sign. The node is not the seat of the oscillation. At the node the force is undefined, and the path below stays where  $u$  is finite.

The outward force on one electron of a symmetric shell is

$$F_{\text{out}}(r) = \frac{ke^2}{r^2} u(x), \quad x = \frac{\ell}{2r}, \quad (21)$$

with  $u$  the combination of nuclear and chord terms already used for the diameter and for the dodecahedron. At a simple root  $u(x_*) = 0$  with  $u'(x_*) > 0$ ,

$$F'_{\text{out}}(r_*) = -\frac{ke^2}{\ell^3} K, \quad K = 8x_*^4 u'(x_*). \quad (22)$$

The electron mass turns the curvature into a frequency,

$$\omega^2 = \frac{K ke^2}{m_e \ell^3}. \quad (23)$$

Inside this linear reading  $\omega$  is fixed by  $K$ , by  $m_e$ , and by  $\ell$ . It does not grow with the amplitude, and it does not grow with the temperature.

The radial root leaves the angular velocity free. In polar coordinates

$$m_e(\ddot{r} - r\dot{\theta}^2) = F_{\text{out}}(r). \quad (24)$$

The term  $m_e r \dot{\theta}^2$  is the kinematics of a tangential velocity. It is not a new contribution in (3). While the velocity is small the root shifts outward by

$$\delta r = \frac{m_e \langle v_\theta^2 \rangle}{|F'_{\text{out}}(r_*)| r_*}. \quad (25)$$

When the tangential energy equals the energy of a radial amplitude  $A$ , the shift is  $A^2/r_*$ , smaller than  $A$  by the factor  $A/r_*$ . The mean radius remains the electrostatic root. The electron goes around the shell while the radius oscillates across the sign boundary. For the diameter the angle is a maximum, by Section 4, so that going-around is not itself a vibration. The vibration is the radial crossing.

Temperature is the energy in the oscillator. Classical equipartition of one radial mode gives

$$\langle (\delta r)^2 \rangle = \frac{k_B T}{|F'_{\text{out}}(r_*)|} = \frac{k_B T \ell^3}{K k e^2}. \quad (26)$$

The root-mean-square amplitude grows as  $\sqrt{T}$ . That growth is the vigor of the motion. The constants  $k_B$  and, below,  $c$  and  $h$  are laboratory constants. They are not fixed by  $\gamma_s$ .

An accelerated charge radiates. The mean Larmor power of one nonrelativistic electron is

$$P = \frac{2ke^2 \langle a^2 \rangle}{3c^3}, \quad a = -\omega^2 \delta r. \quad (27)$$

With (26),  $\langle a^2 \rangle = \omega^2 k_B T / m_e$ , so  $P$  is proportional to  $T$  at fixed  $\omega$ , and proportional to  $K$  at fixed  $T$ . A hotter occupation of the same root radiates more power at the same frequency. A warm body is a large collection of roots. Its heat rays are this radiation. They are not the line of one root.

The curvatures already in hand are far apart. The charge-8 dodecahedron, the laboratory closure of Section 5, has  $K = 4.556$ . The diameter of charge 1, in the reversed layer, has  $K = 126.45$ . On the square, at fixed protons, the curvature in  $\rho$  is  $18.10 ke^2/\ell^3$ , and (23) applies with that number in place of  $K$ . The diagonal stiffness (16) is carried by the proton reduced mass  $m_p/2$ .

The identification  $a = a_0$  of Section 10, together with the CODATA values [2], converts these curvatures into wavelengths

|                         |                       |                         |
|-------------------------|-----------------------|-------------------------|
| dodecahedron, $Z = 8$ : | 196.8 nm,             | 6.299 eV,               |
| square, electron :      | 98.74 nm,             | 12.56 eV,               |
| square, diagonal :      | 4.451 $\mu\text{m}$ , | 2247 $\text{cm}^{-1}$ . |

The first two are ultraviolet. They are the electron oscillating across the sign boundary. The third is infrared. It is the same square, read with the proton mass: the nuclei move along the diagonal and the electrons keep the side, so the electrons are accelerated at the infrared frequency. The harmonic constant of  $\text{H}_2$  in its ground state is  $4401.21 \text{ cm}^{-1}$  [5]. The diagonal computed here lies lower by a factor 1.96.

At 300 K,  $k_B T = 0.02585 \text{ eV}$ . The electron quanta are two or three orders above  $k_B T$ . A comparison that uses the external Planck constant therefore leaves those oscillators empty in a warm body. The classical amplitude (26) does not know this. On the charge-8 dodecahedron it is  $0.0303 \ell$  at 300 K, a fraction 0.116 of the distance from the root to the chord node, and a fraction 0.030 of the radius. The centrifugal shift is then 0.030 of the amplitude. The same classical amplitude reaches the chord node at  $2.22 \times 10^4 \text{ K}$ .

The classical power favors the stiffer root, because  $P$  grows with  $K$ . A warm body does the opposite: its heat rays are infrared, and the ultraviolet shell stays dark. The diagonal quantum is  $0.2785 \text{ eV}$ , which is  $10.8 k_B T$  at 300 K, so that mode is only weakly occupied, and its wavelength is already in the infrared. The laboratory displacement law puts the peak of a body at 300 K at  $9.659 \mu\text{m}$  [2], longer than the diagonal by a factor 2.17. The intermolecular residue at eight side lengths is 0.0088 of the covalent diagonal. A restoring curvature of that order would lie further into the infrared. The parallel residue has no root, and no such curvature is computed here.

The frequency in the Larmor formula is fixed by the sign boundary before a temperature is assigned. With the inertia of the moving coordinate in place of  $m_e$ , (23) is the whole of it: the dimensionless curvature of the root, that inertia, and  $\ell$ . Temperature is the energy with which the root is occupied. It determines the amplitude (26), and through  $a = -\omega^2 \delta r$  the power (27), at a frequency the occupation does not return.

One signed balance can still radiate at two frequencies, because two coordinates can move on it. On the square the cylindrical motion of an electron at fixed protons has curvature  $18.10 ke^2/\ell^3$  and inertia

$m_e$ . The diagonal has stiffness  $8.178 ke^2/\ell^3$  and inertia  $m_p/2$ . By (23) the ratio of the wavelengths is the square root of the ratio of  $K/m$  on the electron coordinate to  $K/m$  on the diagonal. The wavelengths already computed stand in the ratio 45.08. Both scale as  $\ell^{3/2}$ , so the ratio does not use the identification of the side with  $a_0$ . The passage from the ultraviolet root to the infrared root is this square root. The electron radiates in the infrared because the side is constrained to the diagonal: the charge is accelerated at the nuclear period, while its own crossing of the sign boundary remains the ultraviolet root.

Temperature moves the occupation along the order those roots already form. The order is the order of  $K/m$ . At 300 K the displacement peak lies at 0.1283 eV, which is  $4.96 k_B T$ , longer than the diagonal by the factor 2.17 already found, and below the electron quanta by the factors 49 and 98. Raising the temperature raises  $k_B T$ , and the highest quantum the body can sustain therefore moves toward larger  $K/m$ , toward shorter wavelength. The vigor that belongs to ordinary temperature is the slow end of the order, the soft boundary. The electron shares that end only because the side follows the diagonal. Wien's law is the laboratory statement of the motion up the order. It is not a further root of  $\gamma_s$ .

A transition has no such motion inside its frequency. The condition

$$\nu = \frac{E_i - E_j}{h} \quad (28)$$

takes  $\nu$  from a pair of eigenvalues. The eigenvalues are the spectrum of a Hamiltonian. Temperature parameterizes the ensemble over that spectrum and leaves the spectrum unchanged, so the difference is the same in a cool body and in a hot one. What changes is the rate, through the occupation of the upper state. The electron differences computed here, 6.299 eV and 12.56 eV, are invariant in that sense. The diagonal quantum 0.2785 eV is invariant in the same sense, and stands beside the harmonic constant of  $H_2$  as a fixed infrared line. The peak of the heat ray is a different object. It sits at  $4.96 k_B T$  and scales with the temperature. To identify the peak with a transition is to require one frequency to be both an eigenvalue difference and proportional to  $T$ . The spectrum of a fixed Hamiltonian admits no such proportion. That is the contradiction.

The Planck factor  $1/(e^{h\nu/k_B T} - 1)$  places the temperature in an occupation. The frequency of the mode is given first, and the factor then removes every quantum larger than a few  $k_B T$ . Under that removal the electron differences stay at 6.299 eV and 12.56 eV. The peak that moves belongs to a continuum of modes supplied apart from the frequency condition (28).

The restoring roots are that continuum for the signed force, and they are roots of the same  $\gamma_s$  that defines the orbits. Each keeps the frequency (23) of the mass that traverses it. Ordinary temperature occupies the heavy end, and the field it radiates is the Larmor field of the charge accelerated at that end. A hotter body reaches a root of larger  $K/m$ . The characteristic period shortens. Each root stays at the frequency its sign boundary assigned. A transition reading shortens the characteristic period only by taking the frequency off the pair of states and laying an extra factor over a continuum the states did not provide. In the signed-force reading the temperature selects how far the occupation extends along that order. The advantage is that the selection never rescales a frequency. The characteristic period is the period of the root at the frontier of the occupation. Every heavier root keeps the frequency its sign boundary assigned. Once the external quantum is admitted, every lighter root stays empty. Wien's displacement, as a formula for the peak, is not derived.

## 12 Conclusions

The sign of  $\gamma_s$  is the repulsion in (3). A centrifugal term is not required, and it is not introduced. One electron and one proton have no force-free radius. A diameter and a cube about one nucleus each have one restoring radial balance in the first repulsive shell, the diameter is an angular maximum, and 18 counts two chiral seats through the quadrupole rather than a node. For every nuclear charge  $Z \geq 3$  the cube has no root in the outer well. A regular dodecahedron does: for every  $Z \geq 8$  its response equals the charge before the nearest chord meets the first node, and the removal to infinity stays in the outer well. Two of each have exactly one square in the first repulsive shell, with phase between  $3/2$  and  $8/5$ , diagonal force  $4.3532 ke^2/\ell^2$  and stiffness  $8.178 ke^2/\ell^3$ . The bonds that carry  $2/8$  and  $2/12$  of an electron pair have electrostatic roots and indefinite Hessians; their partial forces are 0.04323 and 0.1704 of the covalent diagonal. Neutral pairs separated by eight side lengths retain a parallel force 0.0088 of that diagonal, and at four side lengths an inward pair enters the repulsive shell and the force changes by the sign of  $\gamma_s$ .

The square is a local minimum, and its electrons sit on opposite vertices. The gravitational channel supplies the complementary stability: two electron masses driven inside the first gravitational node meet a

simple pole, so a shared seat is expelled. The factor  $Gm_e^2/(ke^2) = 2.401 \times 10^{-43}$  thins that neighborhood and does not remove it. The node still prevents the outer-well integral from assigning a dissociation energy, and the Bohr spectrum is not implied. One identification, the side equal to  $a_0$ , converts the diagonal into  $\sqrt{2}a_0$ .

A restoring radius is a sign boundary. The radial oscillation across it has  $\omega^2 = Kke^2/(m_e\ell^3)$ , with  $K = 4.556$  on the charge-8 dodecahedron. With the side equal to  $a_0$  that wavelength is 196.8 nm. The electron on the square has wavelength 98.74 nm. The diagonal, carried by  $m_p/2$ , has wavelength 4.451  $\mu\text{m}$ . The ratio of the last two wavelengths is 45.08, the square root of the ratio of  $K/m$  on the electron coordinate to  $K/m$  on the diagonal, and the identification of the side is not required for the ratio. The heat ray is the acceleration of the charge at the period of the root that ordinary temperature occupies. The ultraviolet members stay empty once the external quantum is admitted. A transition frequency is an eigenvalue difference and does not scale with temperature. The peak of a body at 300 K lies at 0.1283 eV and scales with  $T$ . The signed force carries the temperature as the extent of occupation along the order of  $K/m$ . The advantage is that the extent never rescales a frequency: the characteristic period is the period at the frontier of that occupation. Wien's law is not derived.

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